Paleomagnetism and Geochemistry of Some Dikes in Mysore State, India[1]

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Paleomagnetism of 91 samples (370 specimens) and petrochemical analyses of 37 samples from 10 different dikes of Mysore state, India, indicate that these dikes belong to two different groups. The poles of four dikes correspond with the pole position of Deccan traps of Cretaceous-Tertiary age that were presumably erupted when India was in the southern hemisphere. The pole positions of two of the dikes, however, suggest that at the time of their intrusion India was probably in the northern hemisphere. Petrochemically, the former are tholeiitic like the Deccan traps, whereas the latter are alkalic basalts.

In the peninsular Indian shield, there are several dike swarms of different ages and petrographic types intrusive into the Precambrians. *Holland* [1897], *Rama Rao* [1940], and *Pascoe* [1950] considered some of the basic dikes in Indian shield to be of Cuddapah age (Upper Proterozoic), whereas *Pichamuthu* [1959] and *Radhakrishna* [1967] think that some of them probably belong to Deccan trap activity (Upper Cretaceous to Eocene). Petrographically, *Pichamuthu* [1959] has classified Mysore dikes into dolerites, quartz dolerites, olivine dolerites, and epidiorites, including metabasalts and metadiorites.

These earlier studies were based mainly on petrological similarities. In recent years, however, remanent magnetism, chemical composition, and radiometric dating are being utilized in correlating and classifying dike swarms in different parts of the world [*Strangway*, 1961; *Fahrig et al.*, 1965; *Evans*, 1968; *Larson and Strangway*, 1969]. In view of this, a group of ten dikes from Mysore state, India, were subjected to paleomagnetic and petrochemical analyses. The results of this study, presented here, have brought to light the significance of such studies, particularly in the absence of radiometric age data, in classifying dike swarms in shield areas.

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Sampling and Measurements

Despite the handicap of poor exposures in Mysore state, 91 oriented hand samples could be collected from 10 different dikes whose locations are shown in Figure 1. In order to avoid heavy oxidizing conditions prevalent in Mysore state, the samples were collected from road cuttings, quarries, and river beds. From each sample four to six cores were drilled, and finally 370 specimens were selected for paleomagnetic measurements.

Directions and intensities of natural remanent magnetization (NRM) were measured on an astatic magnetometer [*Blackett*, 1952] whose sensitivity is of the order of 3×10^{-7} gauss/cm. The intensities of NRM were found to be spread over a wide range from 0.4×10^{-4} to 206.3×10^{-4} emu/cc. The NRM directions from dikes 1, 2, 4, 5, 6, 7, 8, 9, and 10 showed a large scatter, whereas the NRM directions from dike 3 were clustered. The different parameters of the NRM of each dike are presented in the first part of Table 1.

Alternating Field Demagnetization

The samples from all dikes were subjected to alternating field demagnetization in progressively increasing peak fields of 25, 50, 75, 100, 200, and 300 gauss in an apparatus similar to *Creer's* [1959]. In dikes where the NRM directions were not random, but still somewhat scattered, the directions of magnetization generally reached a stable end point in low fields. The field was, however, found to be high enough to remove the secondary and viscous

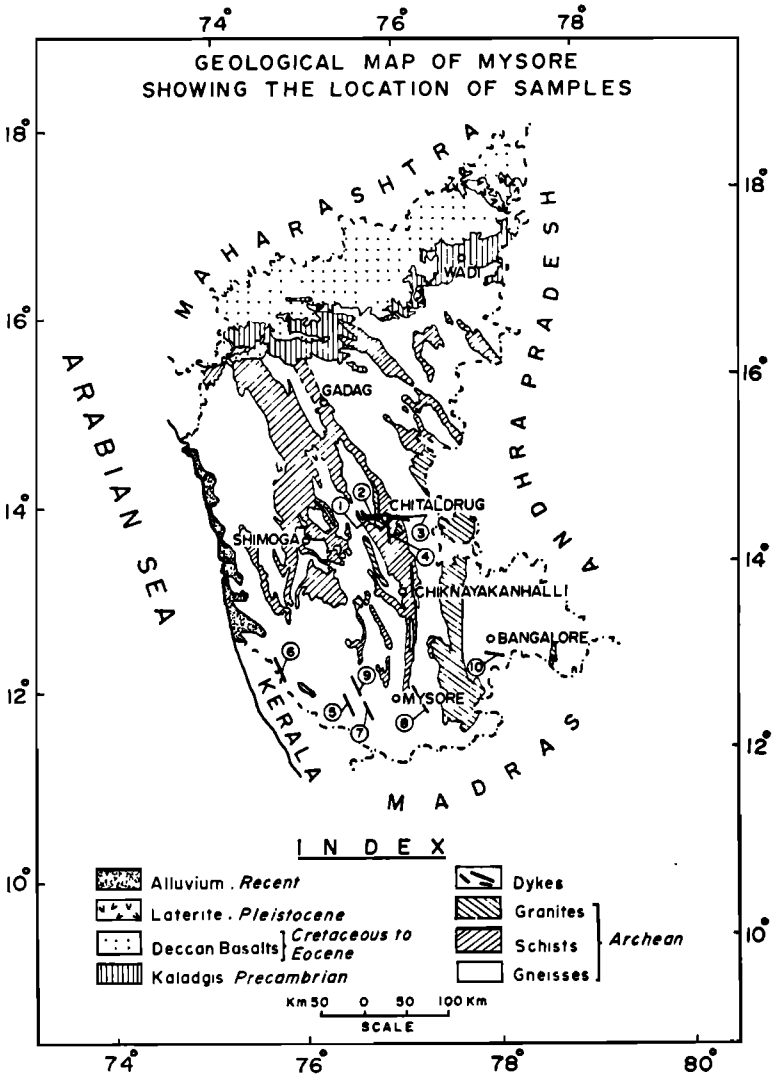


Fig. 1. Geological map of Mysore state showing the location of dike samples.

components of magnetization, indicating the presence of a stable component of magnetization.

The stable components in dikes 1, 2, 5, 6, 7, and 10 were revealed after treatment with low ac fields of the order of 150, 200, 200, 25, 50, and 150 gauss, respectively. All of them eventually yielded well-grouped magnetic directions. The results obtained after the ac demagnetization along with other parameters for the stable dikes are given in the second part of Table 1, and the stable magnetic directions obtained are shown in Figure 2. In general, after ac cleaning, these dikes give precision parameter (K) of 23 or more and have half-angle for the cone of 95% confidence (α_{95}) of less than 12.

However, dikes 3, 4, 8, and 9 failed to give consistent and well grouped directions of magnetization after ac cleaning, even in the high fields. Thus these dikes fall into two categories; magnetically stable and unstable, as is consistent with the findings of *Bruckshaw and Robertson* [1949], *Graham* [1953], and *Larson and Strangway* [1969].

Shown in Figures 3*a*, *b* and 4*a*, *b* are normalized ac demagnetization curves and continuous

TABLE 1. Site Mean Directions for the Mysore Dikes

Site Location	Lat.	Long.	N	E	n	Before Cleaning, NRM D	I	R	K	After Cleaning D	I	R	K	α_{95}	λ	VGP λ_p	L_p	δ_m	δ_p
1	14°10′	76°27′	12	—	45	359	−12	10 82	9 8	2	−30	11.57	25 5	8 6	21	55°N	106°W	9 6	6 8
2	14°11′	76°22′	13	1	48	12	+45	10 77	8 25	26	+50	11.70	36 7	12 8	30	45°N	138°E	17 2	11.4
3	14°15′	76°25′	4	—	11	6	+43	3.96	75 00	Random directions				—	—	—	—	—	—
4	14°10′	76°24′	8	—	22	—	—	—	—	Random directions				—	—	—	—	—	—
5	12°11′	76°04′	5	—	40	—	—	—	—	8	−59	4 91	44.4	9.9	39	39°N	112°W	14.9	10.8
6	12°25′	75°11′	13	4	56	359	−55	7 28	4 06	18	−62	8 85	53.3	6 7	43	32°N	118°W	8.5	5.3
7	12°10′	76°14′	22	2	82	325	−14	8.18	1 52	329	−48	19 20	23.8	6 5	28	39°N	68°W	10.4	8 0
8	12°18′	76°50′	7	—	21	341	+47	4 95	2.05	Random directions				—	—	—	—	—	—
9	12°20′	76°11′	3	—	23	—	—	—	—	Random directions				—	—	—	—	—	—
10	12°44′	77°31′	4	—	22	238	−5	2 07	1 56	141	−75	3.94	50.0	9.9	61	34°N	55°E	18 5	16.3

Various specimens exhibited stable RM directions at various peak fields ranging from 50 to 300 oe.

N total number of samples.
E number of samples rejected.
n number of specimens considered.
D declination of RM in degrees.
I inclination of RM in degrees: down if positive, up if negative.
R length of resultant *N* unit vector.
K precision parameter.
α_{95} semi-angle of the cone of 95% confidence.
VGP virtual geomagnetic pole.
λ ancient latitude of dike.
λ_p, L_p latitude and longitude of VGP.
δ_m, δ_p semi-axes of the oval of confidence about the poles of 95 probability level.

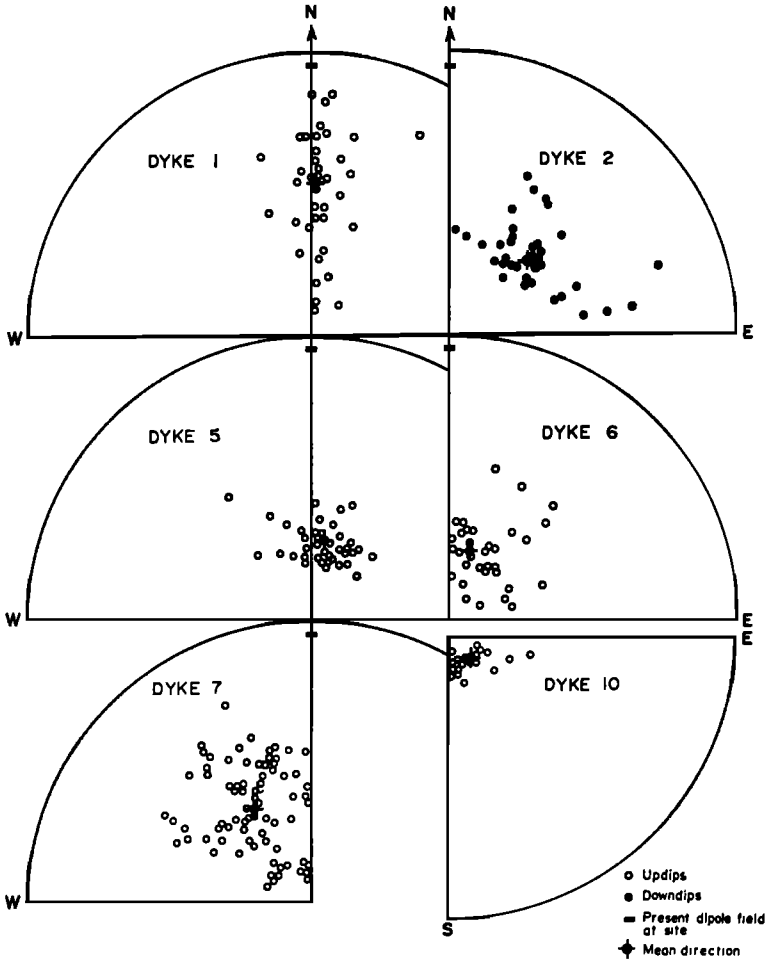


Fig. 2. Stereographic projection of magnetically cleaned directions of magnetization from dikes 1, 2, 5, 6, 7, and 10.

changes in direction of magnetization for dikes 1 and 2. The specimens have reached stable directions in a field ranging from 50 to 300 gauss. However, the vectors of both the dikes show stability to the progressive ac demagnetization. As a further check, a few ac-cleaned specimens from dikes 1 and 2 were heated to 300°C to test their thermal stability. Within the limits of experimental error, they remained in the group that was formed by ac demagnetization, thus indicating the truly NRM nature of the magnetization. Similar stability tests from all the six dikes suggest NRM in Mysore dikes to be of TRM nature.

Virtual geomagnetic poles were computed assuming geocentric axial dipole field. Figure 5 shows the plot of pole positions of the stable magnetized dikes (1, 2, 5, 6, 7, and 10) and of Upper and Lower Deccan traps, along with the ovals of confidence at 95% probability level. It can be seen from the figure that the poles obtained from the dikes 1, 5, and 6 are close to each other, suggesting their similar magnetization age. The poles of these dikes and that of dike 7 are in close agreement with the mean pole positions calculated for Upper and Lower Deccan traps by *Athavale et al.* [1968]. A very close agreement in pole positions between the two formations separated by a distance of about 1000 km also indicates that the geomagnetic field must have been predominantly an axial dipole. The poles of dikes 2 and

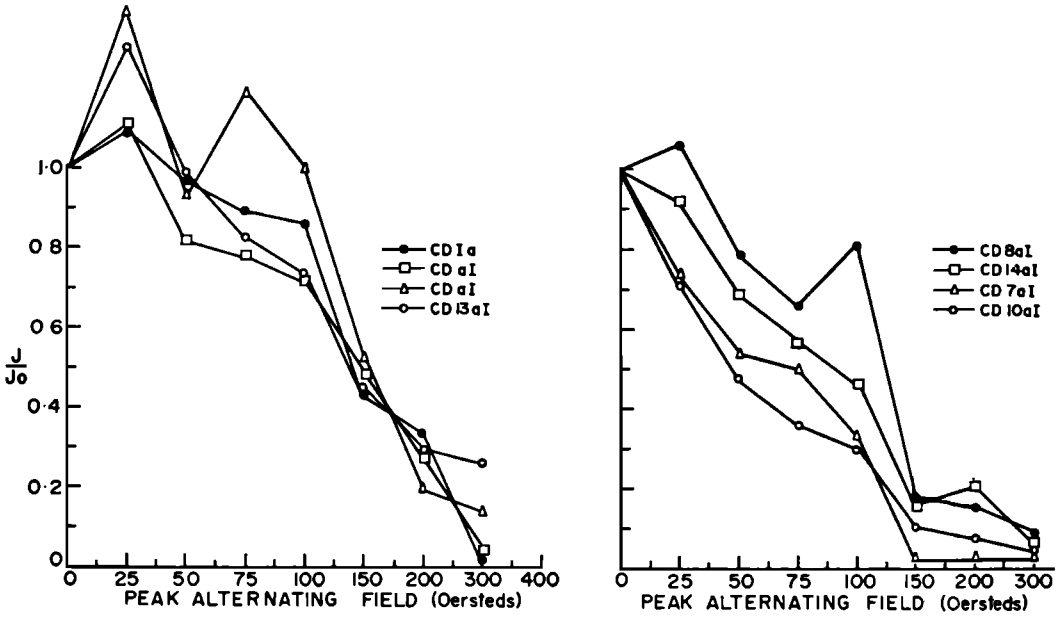


Fig. 3*a*. Normalized ac demagnetization curves for typical specimens from dike 1.

10 are not similar to any other paleomagnetic pole of India. However, they suggest the position of India to be in the northern hemisphere.

Chemistry of the Dikes

Representative samples from all dikes have been chemically analyzed by rapid silicate methods [*Shapiro and Brannock*, 1962].

The data on Mysore dikes, along with those for average Deccan basalts [*Washington*, 1922], average tholeiite dolerites, and average alkali dolerites [*Manson*, 1967], are given in Table 2. Alkali-silica ratios (Figure 6) of dikes 2, 4, and 10 place them in the alkali basalt field as defined by *MacDonald and Katsura* [1964], and the rest are placed in the tholeiitic field. In addition, high titanium and low silica are also present in the three alkali field dikes, a feature that distinguishes alkali from tholeiite basalts. A comparison of chemical composition

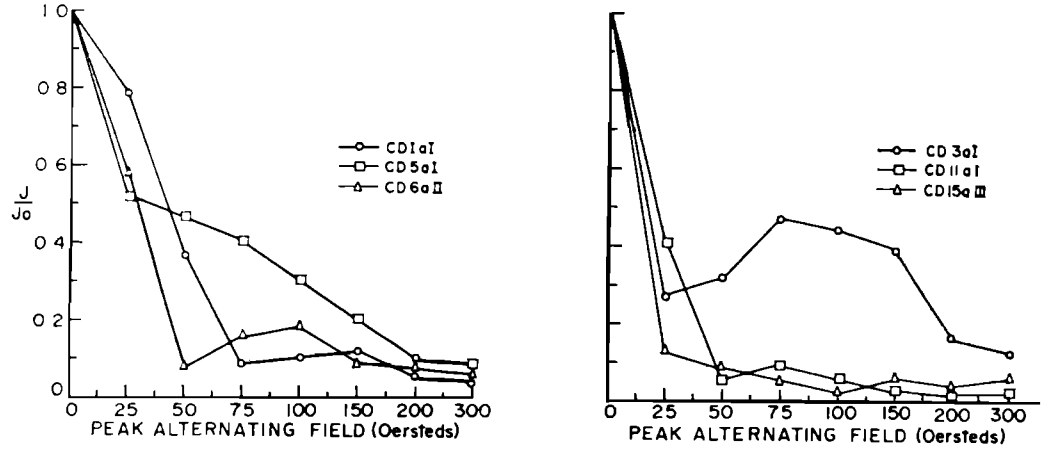


Fig. 3*b*. Normalized ac demagnetization curves for typical specimens from dike 2.

of dikes 1, 3, 5, 6, 7, 8, and 9 with those of the magmas from the Deccan traps, average of tholeiite dolerites, and alkali dolerites (Table 2) also shows a similarity with the Deccan traps and tholeiite dolerites. Yet another point of similarity between these dikes and Deccan traps is the enrichment of iron, caused by fractionation of pyroxenes.

Discussion of Results

From the data presented here, it should appear that the dikes belong to two distinct groups. The poles of dikes 1, 5, 6, and 7 indicate the position of India to be in the southern hemisphere and are similar in position to those of Deccan traps. Petrochemically, these dikes and dikes 1, 3, 8, and 9 are typical tholeiite. If this inference should hold true, then it could be argued that the intrusion of these and similar dikes in the peninsular shield at the time of Deccan traps activity took place in tensional cracks freshly produced by the northward drift of India or some other causes. Alternatively, however, it could also be a simple case of filling up of ancient zones of weakness, perhaps opened up by the impact of the Himalayan orogeny, somewhat similar to the intrusion of Spanish peak dike swarms of Colorado, which filled the Precambrian fracture zones at the time of the Laramide orogeny [*Badgley*, 1960].

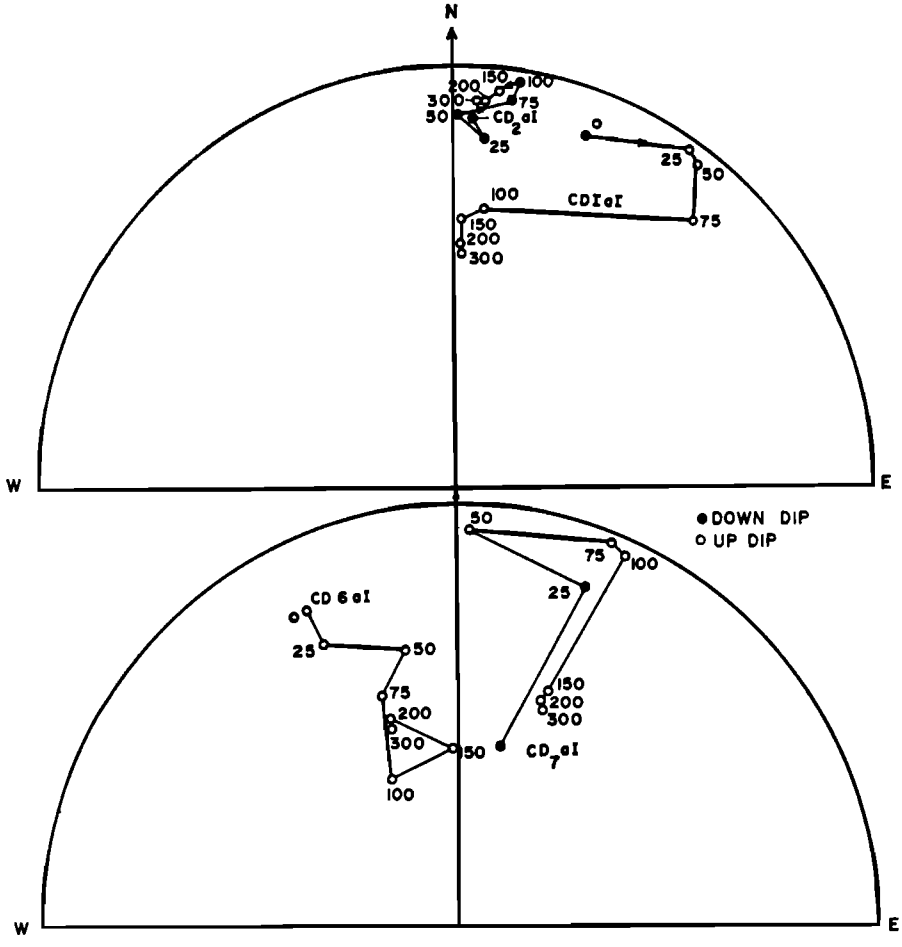


Fig. 4*a*. Equal area projection showing changes in direction of magnetization after cleaning in various peak alternating fields for typical specimens from dike 1.

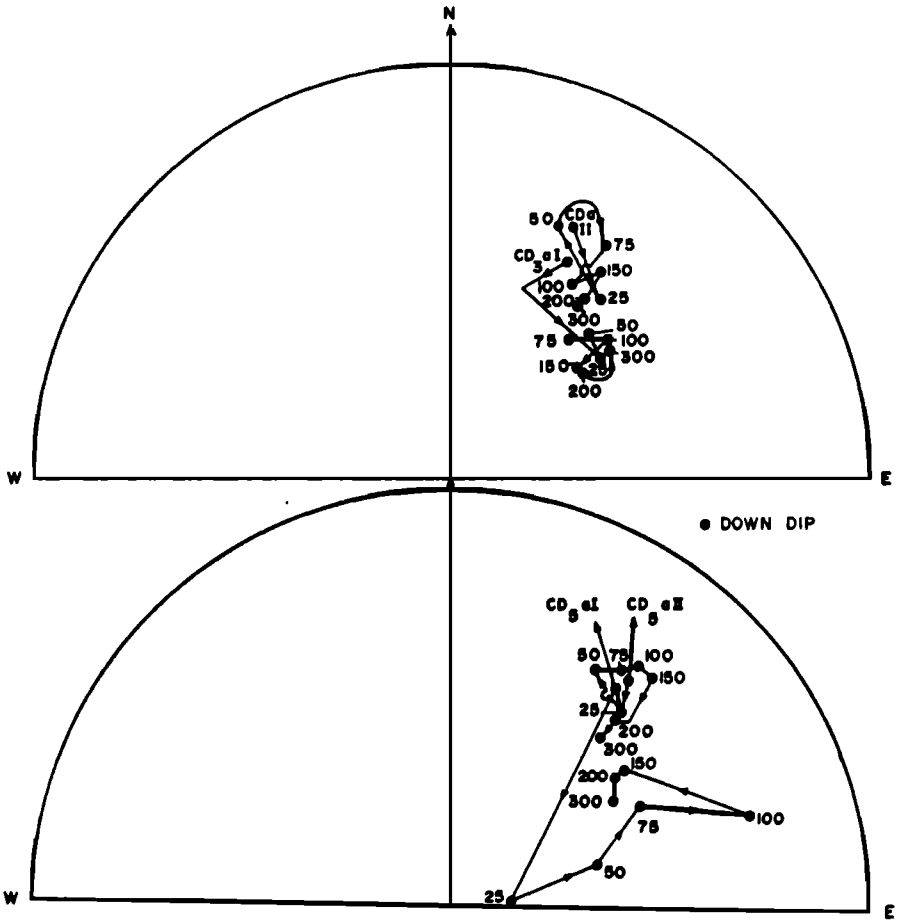


Fig. 4*b*. Equal area projection showing changes in direction of magnetization after cleaning in various peak alternating fields for typical specimens from dike 2.

TABLE 2. Chemical Composition of Mysore and Comparable Rocks from the Literature

Component	Mysore Samples												
	1	2	3	4	5	6	7	8	9	10	11*	12†	13‡
SiO_2	50.25	48.79	50.17	49.09	50.76	49.20	50.37	50.74	50.84	48.57	50.61	51.20	48.30
Al_2O_3	17.54	17.02	14.21	13.84	13.66	15.52	16.13	14.79	13.28	11.16	13.58	15.10	16.00
Fe_2O_3	3.10	4.12	3.05	3.50	2.64	3.40	3.01	3.28	3.50	2.8	3.91	2.3	2.8
FeO	11.33	10.39	11.77	11.80	10.29	11.10	11.32	11.04	10.70	10.90	9.92	8.9	8.3
CaO	5.20	9.93	8.01	8.03	9.5	8.80	8.50	10.90	12.30	11.40	9.45	10.10	2.8
MgO	2.97	3.49	3.35	3.53	1.85	3.60	3.55	3.02	2.70	3.10	5.46	6.40	6.50
Na_2O	2.45	2.73	2.45	3.30	2.45	2.70	2.50	2.50	2.20	3.00	2.60	2.20	3.30
K_2O	1.12	0.51	0.60	0.49	0.80	0.43	0.47	0.51	0.52	0.77	0.72	0.90	1.30
TiO_2	1.95	2.15	1.84	2.89	1.72	1.26	0.51	1.46	1.24	2.72	1.91	1.50	1.80
MnO	0.20	0.19	0.13	0.19	0.12	0.12	0.65	0.20	0.19	0.12	0.16	0.19	0.17
P_2O_5	0.34	0.36	0.38	0.39	0.36	0.45	0.41	0.31	0.43	0.59	0.39	0.20	0.32
H_2O	—	—	—	—	—	—	—	—	—	—	—	0.90	1.40

* The average of Deccan trap, after *Washington* [1922].
† Averages of all tholeiitic dolerites, after *Manson* [1967].
‡ Averages of all alkalic dolerites after *Manson* [1967].

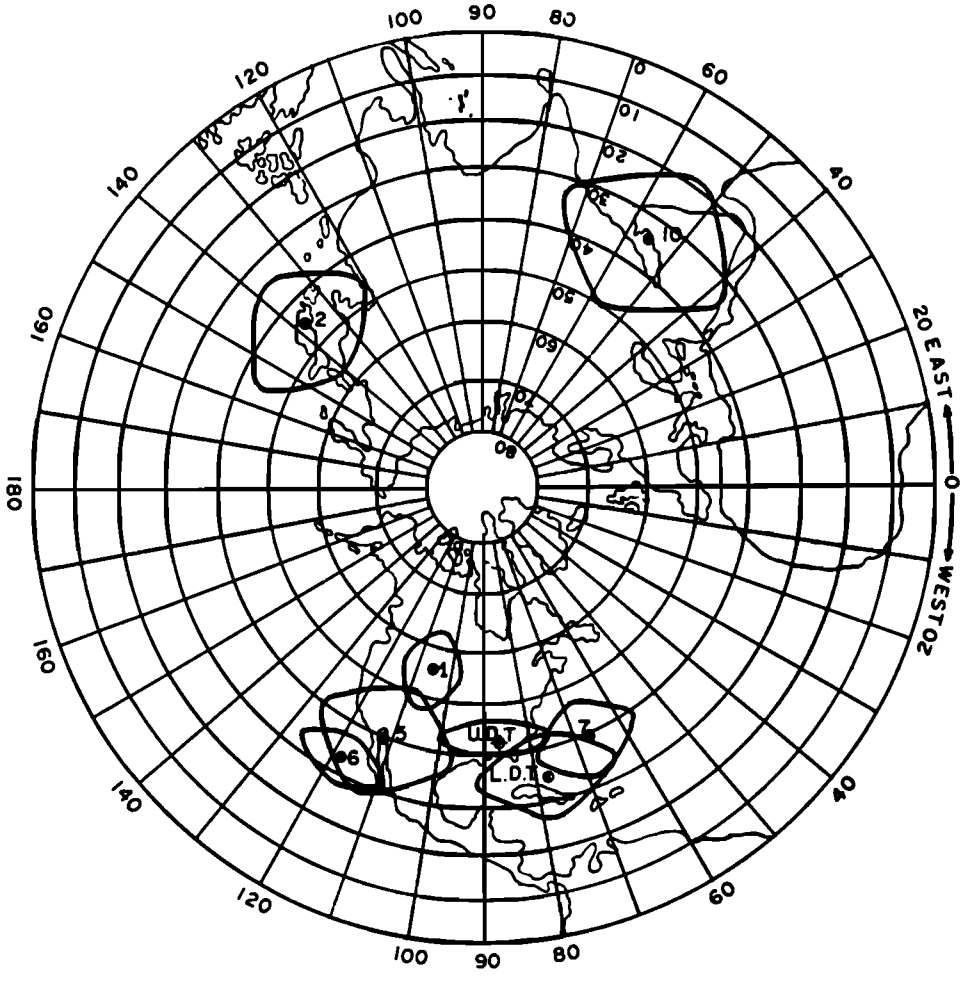


Fig. 5. Virtual geomagnetic pole positions of six Mysore dikes (1, 2, 5, 6, 7, and 10) compared with those of mean Upper Deccan trap pole position (U.D.T.) and Lower Deccan trap pole position (L.D.T.).

The poles of dikes 2 and 10 suggest the position of India to be in the northern hemisphere, and petrochemically these dikes and dike 4 are alkalic basalts. It is relevant to mention that paleomagnetic studies of some of the Precambrian rocks of India [*Athavale et al.*, 1963] have shown that, during parts of the Precambrian, India was in the northern hemisphere. Hence it can be inferred that these dikes were probably intruded during the Precambrian when India was in the northern hemisphere.

An interesting aspect of the present study is the probable occurrence of alkali basalt of Precambrian age, as until lately it was thought that the continental and sill swarms were the intrusive equivalents of tholeiite flood basalt [*Turner and Verhoogen*, 1960, p. 209]. Similar occurrences of alkali basalts of radiometrically dated Precambrians have been reported in recent years [*Fahrig et al.*, 1965; *Condie et al.*, 1969]. Thus the generalization that all alkali basalt intrusions are post Precambrian may not appear to be true.

Although the present study is limited in that only ten dikes have been analyzed, it does seem to bring out the importance of such studies in shield areas for classifying and correlating dike swarms.

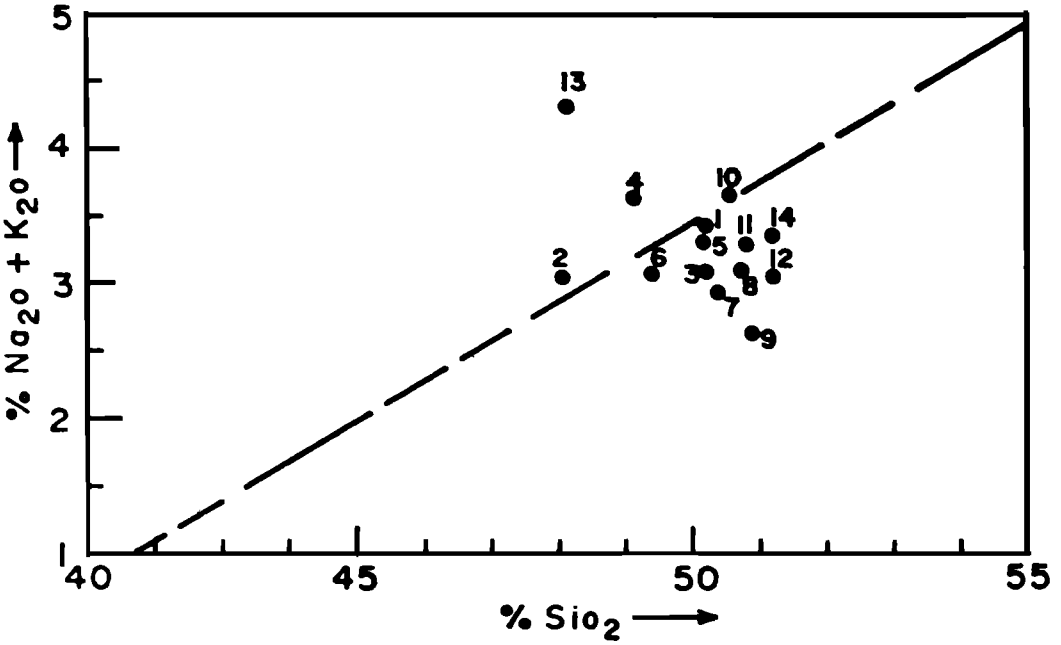


Fig. 6. Alkali-silica ratio diagram of ten Mysore dikes (1–10); numbers 11, 12, and 13 represent comparable rock types from Table 2.

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